10BONDS

# Livio Rating (LR) — Corporate Credit Risk Methodology

Proposed institutional-grade framework · Version 2.0 · 19 October 2026

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# Overview

The methodology should go substantially beyond conventional financial ratio analysis. The objective is to assess not only whether a company can repay its debt under current conditions, but whether it can continue to service and repay that debt when its business model, management, supply chain, financing or operating environment is disrupted.

I would build the methodology around three fundamental questions:

1. Financial resilience: Can the company service and repay its obligations based on its current financial position and expected cash generation?
2. Structural resilience: Will the company remain competitive and economically viable over the life of the bond?
3. Shock resilience: Can the company survive plausible adverse events, including geopolitical conflict, financial crises, commodity shocks, technological disruption and management failures?

The important distinction is between risk exposure, vulnerability and actual financial impact. For example, an increase in oil prices can be harmful to an airline, beneficial to an oil producer and relatively neutral to a software company. The methodology must capture these differences quantitatively wherever possible.

A credible LR framework should therefore combine financial metrics, forward-looking business analysis, event-risk assessment and explicit stress testing, with a transparent separation between observable facts and analytical judgments.

# 1. Architecture

The published framework should distinguish between the base credit score, the financial consequences of adverse scenarios, and exceptional credit events.

**A. Base credit assessment** LR Base · 0–100
Financial strength + business resilience + management + structural exposures

**B. Scenario engine**
Recalculate cash flows, liquidity and leverage under defined shocks

**C. Event-risk overlay**
Governance failures, litigation, extraordinary transactions and other discrete events

**D. Final corporate credit assessment** LR Final · 0–100
With separate scenario results, confidence indicators and critical-risk flags

This is directionally consistent with established rating-agency practice, which distinguishes business and financial risk, modifiers and group support, evaluates financial policy and event risk, and separates foreseeable vulnerabilities from unforeseen event risks.

LR's proposed differentiation: a standardized, transparent, issuer-specific shock-transmission model that makes the consequences of external risks visible to bond investors rather than relying only on a broad qualitative risk judgment.

# 2. Base score

I recommend the following starting allocation for nonfinancial corporates.

**A. Financial strength** **35%**

Leverage 9%; interest coverage 6%; FFO/debt 6%; FCF generation 6%; profitability and volatility 4%; leverage trajectory 4%

**B. Liquidity and repayment** **20%**

24-month liquidity 7%; maturity profile 5%; refinancing flexibility 4%; covenants and contingent obligations 4%

**C. Business model resilience** **20%**

Competitive position 5%; industry outlook 4%; customer/product concentration 3%; technological and China-related disruption 4%; geographic and supplier diversification 4%

**D. Management and governance** **10%**

Governance 3%; financial policy 3%; key-person dependency and succession 2%; execution and risk management 2%

**E. Structural and external exposure** **15%**

Geopolitical 4%; commodities and energy 3%; financial-system sensitivity 3%; physical and climate exposure 2%; legal/regulatory 3%

These weights are proposed research hypotheses, not empirically established optimal weights. Before publication as a validated rating methodology, they must be tested against historical defaults, rating migrations and observed financial deterioration.

A crucial rule: factors in Pillar E measure persistent exposure and vulnerability. The scenario engine measures the incremental consequences of shocks. We must not deduct the same risk twice.

# 3. Risk universe

The following taxonomy is designed as the master risk register. Each risk receives a materiality assessment, evidence record and defined transmission path into financial performance.

## A. Geopolitical

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **Interstate war** | Operations, assets, employees or customers in conflict zones | Production losses, asset impairment |
| **Regional military escalation** | Exposure to Taiwan Strait, Middle East, Eastern Europe and other flashpoints | Supply interruption, logistics and demand |
| **Trade wars** | Tariff exposure by product and destination | Margin compression, lost volumes |
| **China-driven disintermediation** | Ability of Chinese manufacturers to replace the issuer's products or customer relationships | Market-share loss, lower prices |
| **Chinese technology competition** | Relative cost, quality, scale and innovation | Structural EBITDA decline |
| **Sanctions** | Counterparties, countries, technologies and currencies | Inability to trade, stranded assets |
| **Export controls** | Dependence on restricted chips, software or equipment | Production and revenue disruption |
| **Nationalization/expropriation** | Physical assets in higher-risk jurisdictions | Asset loss |
| **Political instability** | Civil unrest, regime change, institutional deterioration | Operational shutdown, higher costs |
| **Strategic shipping chokepoints** | Taiwan Strait, Strait of Hormuz, Suez, Panama | Freight, fuel and inventory costs |
| **Capital and currency controls** | Ability to repatriate cash or service foreign debt | Debt-payment constraints |
| **Protectionism/localization** | Requirements for local manufacturing or procurement | Additional capex, reduced profitability |

China risk must be measured at the product and market level, not assigned mechanically based on the issuer's nationality. For example, a European industrial company could have limited sales in China but substantial exposure to Chinese competitors in Europe. Its competitive risk could therefore be high despite low direct Chinese revenue.

## B. Economic and financial

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **Global recession** | GDP sensitivity of demand | Revenue and EBITDA decline |
| **Financial crisis** | Funding availability and credit spreads | Refinancing failure |
| **Banking-system crisis** | Deposit concentration and reliance on bank facilities | Liquidity constraints |

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **Interest-rate increases** | Floating-rate debt, refinancing calendar | Higher interest expense |
| **Inflation** | Ability to pass through labor and input costs | Margin deterioration |
| **Deflation** | Pricing power and fixed debt obligations | Falling cash generation |
| **Currency depreciation** | Currency mismatch between cash flows and debt | Higher debt-service burden |
| **Sovereign default** | Exposure to sovereign counterparties, banking and capital controls | Receivables losses, financing disruption |
| **Commercial real-estate crisis** | Tenant, collateral and customer exposures | Impairment, demand decline |
| **Credit-market closure** | Debt maturities during inaccessible markets | Funding gap |
| **Counterparty default** | Concentrated receivables and hedging counterparties | Cash losses |
| **Pension funding crisis** | Defined-benefit deficits and mandatory contributions | Increased cash obligations |

### C. Commodities and energy

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **Oil price spike** | Fuel intensity, hedging, price pass-through | Higher costs or revenue |
| **Oil price collapse** | Upstream production and energy-linked demand | Lower revenue |
| **Natural gas disruption** | Gas dependence, contracted supply | Higher costs, shutdown |
| **Electricity price surge** | Power intensity and electricity contracts | EBITDA reduction |
| **Critical mineral shortages** | Copper, lithium, rare earths and other essential inputs | Production constraints |
| **Agricultural commodity shocks** | Food-processing and agricultural exposure | Input inflation |
| **Water scarcity** | Water-intensive facilities and geographic exposure | Lower capacity, capex |
| **Carbon pricing** | Emissions intensity and regulatory exposure | Higher compliance costs |
| **Energy transition** | Stranded assets and changes in end-market demand | Long-term earnings erosion |

Every commodity shock must be assessed for its direction, magnitude, hedging protection, pass-through capacity and time lag.

### D. Technology

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **AI substitution** | Proportion of products/services that AI could replace | Structural revenue loss |
| **Automation** | Competitors' ability to reduce costs faster | Margin pressure |

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **Platform disintermediation** | Dependence on distributors, marketplaces and aggregators | Loss of customer ownership |
| **Technological obsolescence** | Product life cycle, R&D effectiveness | Market-share loss |
| **Cyberattack/ransomware** | Cybersecurity controls and recovery capabilities | Shutdown, recovery cost |
| **Cloud concentration** | Dependency on one cloud or software provider | Business interruption |
| **Data breach** | Sensitive data, compliance and liability | Fines, litigation, customer losses |
| **Patent expiry** | Revenue protected by intellectual property | Pricing collapse |
| **Product substitution** | Alternative technologies or materials | Declining addressable market |
| **Digital infrastructure failure** | Dependency on payments, connectivity and digital systems | Revenue interruption |

## E. Supply chain

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **Single-source suppliers** | Availability of qualified substitutes | Production stoppage |
| **Geographic concentration** | Production clustered in one country or region | Correlated disruption |
| **Semiconductor shortages** | Critical electronic components | Production reduction |
| **Logistics disruption** | Dependence on shipping lanes and transport modes | Delays, increased costs |
| **Supplier bankruptcy** | Financial strength of critical suppliers | Replacement costs |
| **Manufacturing accidents** | Safety and redundancy of major facilities | Shutdown, liabilities |
| **Quality failure/product recall** | Product safety, traceability and recall exposure | Cash outflow, brand damage |
| **Labor strikes** | Union concentration, labor relations | Lost production |
| **Skilled labor shortages** | Availability of essential technical capabilities | Capacity and cost pressures |
| **Infrastructure failures** | Grid, water, telecom and transport dependencies | Operating interruption |
| **Inventory disruption** | Just-in-time exposure, safety stock | Working-capital shock |

## F. Health and nature

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **Global pandemic** | Business shutdown sensitivity and demand shifts | Revenue, labor and liquidity |
| **Regional epidemic** | Geographic concentration and workforce exposure | Local interruption |
| **Earthquakes** | Facilities in seismic zones | Physical losses |
| **Flooding** | Asset locations and business continuity | Damage and downtime |
| **Hurricanes/typhoons** | Production, distribution and supplier exposure | Disruption and repair costs |
| **Extreme heat** | Labor productivity, cooling and water requirements | Higher operating costs |

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **Wildfires** | Facilities and logistics corridors | Shutdown and asset damage |
| **Drought** | Agricultural, energy and industrial dependencies | Supply constraints |
| **Environmental accidents** | Pollution and remediation exposure | Fines, cleanup, litigation |
| **Insurance withdrawal** | Availability and affordability of catastrophe coverage | Higher retained losses |

## G. Management

This category deserves particular attention because conventional financial statements often fail to capture management concentration and operational fragility.

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **CEO death/incapacity** | Succession plan and delegation of authority | Strategic and operating disruption |
| **Founder dependency** | Critical relationships, knowledge and decisions held by one person | Business continuity risk |
| **CFO departure** | Financial controls and funding relationships | Reporting and financing disruption |
| **Key technical expert loss** | Proprietary knowledge concentrated in individuals | Product/production interruption |
| **Management-team turnover** | Stability of senior leadership | Execution deterioration |
| **Poor succession planning** | Identified and prepared replacements | Extended leadership vacancy |
| **Fraud/accounting manipulation** | Audit findings, restatements and controls | Liquidity crisis, covenant breach |
| **Aggressive acquisitions** | Management's leverage appetite | Debt increase |
| **Shareholder extraction** | Dividends and buybacks funded by debt | Reduced creditor protection |
| **Ownership disputes** | Shareholder litigation and control conflicts | Strategic paralysis |
| **Poor board oversight** | Independence, risk oversight and incentives | Excessive risk-taking |
| **Executive compensation** | Incentives favoring short-term equity returns over creditor interests | Higher financial risk |
| **Succession disputes** | Competing heirs or controlling shareholders | Governance instability |

A key-person event should not automatically cause a major score deduction. A company with a strong succession plan, delegated authority and institutionalized customer relationships may be highly resilient to the loss of an executive. Conversely, a founder-controlled company in which one individual controls technology, funding relationships and customer contracts may have substantial key-person risk.

## H. Legal and strategic

| RISK | WHAT LR MUST INVESTIGATE | FINANCIAL TRANSMISSION |
|---|---|---|
| **Antitrust enforcement** | Market dominance and competition investigations | Fines, divestitures |
| **Major litigation** | Probability and magnitude of adverse judgments | Cash outflow |
| **Product liability** | Potential claims and insurance protection | Liabilities |
| **Regulatory prohibition** | Potential restrictions on core products | Revenue loss |
| **Tax-law changes** | Tax jurisdiction and structure | Higher cash taxes |
| **License revocation** | Dependence on licenses and concessions | Operating interruption |
| **Debt-funded acquisition** | Acquisition size and financing policy | Leverage increase |
| **Leveraged buyout** | Change-of-control provisions and owner incentives | Capital structure deterioration |
| **Corporate restructuring** | Execution and financing requirements | Costs, covenant pressure |
| **Parent-company distress** | Group funding dependence and cash extraction | Liquidity loss |
| **Guarantee withdrawal** | Contractual support and guarantee enforceability | Reduced credit protection |
| **Unexpected capital expenditure** | Compliance, environmental or operational requirements | FCF deterioration |

# 4. From risk to numbers

A risk checklist is useful, but insufficient. A credible model needs to show how an identified risk can impair the company's capacity to repay debt. I propose a five-stage calculation.

**Stage 1 — Exposure.** Determine which portion of revenue, EBITDA, assets, procurement, production or debt is exposed to the event.

**Stage 2 — Vulnerability.** Evaluate the company's ability to mitigate the shock through pricing, hedging, insurance, diversification, substitution or contingency plans.

**Stage 3 — Financial impact.** Translate the shock into changes in revenue, EBITDA, working capital, capital expenditure, free cash flow, interest expense and debt.

**Stage 4 — Timing and recovery.** Model the event's duration, timing relative to debt maturities, recovery path and any permanent deterioration.

**Stage 5 — Repayment consequences.** Recalculate leverage, interest coverage, liquidity and the issuer's ability to meet contractual payments under the stressed scenario.

### Illustrative example: oil price shock

| PARAMETER (HYPOTHETICAL EUROPEAN INDUSTRIAL COMPANY) | BASE CASE | OIL SHOCK |
|---|---|---|
| **Revenue** | €5,000m | €5,000m |
| **EBITDA** | €800m | €680m |
| **Net debt** | €2,000m | €2,000m |
| **Annual cash interest** | €100m | €100m |
| **Net debt / EBITDA** | 2.50× | 2.94× |
| **EBITDA / interest** | 8.00× | 6.80× |

Assume oil rises by 50%, the company experiences €200m in incremental gross operating costs, and it recovers €80m through hedging and customer price increases. EBITDA consequently falls by €120m. The credit assessment must then examine whether the company still generates sufficient free cash flow, whether its debt covenants remain satisfied and whether it can refinance or repay upcoming maturities. The model should also test whether the shock persists for one, two or three years.

# 5. Scenario engine

I recommend a common set of scenarios applied consistently across issuers, with company-specific transmission assumptions.

| SCENARIO | STANDARDIZED SHOCK SPECIFICATION | PRIMARY OUTPUT |
|---|---|---|
| **Global recession** | Revenue volume decline calibrated to industry sensitivity | EBITDA and FCF |
| **Severe financial crisis** | Credit spread increase and temporary market-access closure | Refinancing gap |
| **Oil crisis** | +50% oil price, with a second more severe sensitivity | Net EBITDA impact |
| **Energy crisis** | Material gas/electricity price increase and supply interruption | Operating costs |
| **China competitive shock** | Accelerated price and market-share pressure in exposed products | Structural earnings loss |
| **Taiwan conflict** | Semiconductor and trade disruption, plus regional demand effects | Production and cash flow |
| **Middle East conflict** | Energy and shipping disruption | Costs and logistics |
| **Trade war** | Higher tariffs and market-access restrictions | Margins and volumes |
| **Pandemic** | Demand interruption, absenteeism and operating restrictions | Liquidity runway |
| **Supply chain failure** | Loss of a critical supplier for a defined period | Lost production |
| **Cyberattack** | Temporary system outage and recovery expenditure | Cash losses |
| **Interest-rate shock** | +300 basis points on relevant debt exposure | Interest coverage |
| **Currency crisis** | Material depreciation in exposed currencies | FX losses and debt service |
| **Climate catastrophe** | Loss of a major facility and associated insurance recovery | Liquidity and asset losses |
| **Key-person event** | Sudden loss of a critical executive or technical leader | Execution and revenue effects |
| **Combined systemic crisis** | Recession + financing closure + commodity shock | Maximum credible stress |

These shock magnitudes are illustrative scenario design choices, not estimated forecasts or calibrated tail-event quantiles. They should be refined using historical observations, sector characteristics and empirical testing.

### Scenario impact should not be an arbitrary score

For each scenario s, calculate LR(s) = f(stressed financial and business inputs), then measure ΔLR(s) = LR(base) − LR(s). The model should report both the score change and the underlying financial deterioration.

Manufacturer: LR 78 → 43 under combined stress.

Utility: LR 74 → 66 under combined stress.

This allows an investor to understand that two companies with identical base credit scores may have substantially different vulnerabilities. A manufacturer with LR 78 that falls to 43 under a plausible combined stress may be less resilient than a utility with LR 74 that remains at 66. However, the combined scenario must explicitly model correlated shocks. It cannot be constructed by simply adding the separate losses from every scenario.

# 6. Event likelihood

Not every possible event should receive an invented annual probability. I recommend three classes:

| RISK CLASS | METHODOLOGY | EXAMPLE |
|---|---|---|
| **Structural risk** | Incorporate into base forecast and competitive assessment | Persistent Chinese price competition |
| **Measurable stochastic risk** | Use historical distributions and calibrated scenarios | Interest rates, commodity prices |
| **Discrete event risk** | Apply conditional scenarios and materiality flags; assign probability only with credible evidence | War, CEO death, nationalization |

The important distinction is between a risk that is already present and an event that might occur. For example, reliance on one critical executive is an observable structural vulnerability. The executive's sudden death is a discrete event. The first belongs in the base resilience assessment; the second can be tested as a scenario without inventing a mortality-based credit adjustment.

# 7. Critical-risk overrides

A weighted-average score has an important weakness: a company could score well on most factors while facing one existential risk. The methodology therefore needs override rules. I propose that the following conditions trigger mandatory review rather than simply reducing the score by a few points:

- Insufficient liquidity to meet contractual obligations within the next 12 months under the base case.
- Material debt covenant breach without a documented waiver or credible remedy.
- A near-term refinancing requirement without demonstrated funding access.
- A credible event threatening the company's ability to continue operating.
- Material financial reporting uncertainty or evidence of serious accounting manipulation.
- A material legal, sanctions or regulatory restriction preventing debt service.

These should trigger a documented credit committee review, possible score cap or distressed-credit classification. The precise caps must be specified and empirically tested before production use. This approach prevents an otherwise strong average score from concealing a critical repayment problem.

# 8. Validation

A comprehensive taxonomy alone will not make LR credible to an experienced rating-agency analyst. The credibility will come from how the model handles evidence, uncertainty, financial causality and historical performance. I recommend six mandatory methodological controls.

| CONTROL | REQUIRED STANDARD |
|---|---|
| **Data traceability** | Every material input must have a source, observation date, definition and calculation record |
| **Sector comparability** | Financial thresholds and risk transmission calibrated by industry |
| **No double counting** | Each risk's impact allocated to the appropriate base factor or incremental scenario |
| **Reproducibility** | Same verified inputs and methodology version produce the same score |
| **Historical validation** | Back-test against defaults, distress events and credit deterioration using information available at each historical date |
| **Uncertainty disclosure** | Publish data coverage, confidence, assumptions and unquantifiable risks |

### Validation tests

Before describing LR as a predictive rating model, I would require:

1. Discrimination: Do lower LR scores precede defaults more frequently than higher scores? Test with ROC-AUC and related ranking metrics.
2. Calibration: If the model eventually estimates default probabilities, do predicted probabilities match observed frequencies?
3. Early-warning performance: Does LR identify deterioration 12–36 months before default or restructuring?
4. Stability: Does the score change for economically meaningful reasons rather than minor reporting differences?
5. Incremental value: Do geopolitical, technology and management-risk factors improve predictive performance beyond conventional financial metrics?
6. Benchmark comparison: How does LR perform relative to published agency ratings, accounting-based models and simpler financial-ratio benchmarks?

The fifth test is especially important. It would demonstrate whether the additional complexity of LR produces genuine analytical value.

# 9. Publication

I would position LR as a forward-looking corporate credit resilience methodology, rather than claiming it is more accurate than agency ratings before validation. The public methodology should contain four parts:

| PUBLICATION | CONTENT |
|---|---|
| **1. Corporate LR Methodology** | Objectives, 100-point framework, all factors, definitions, formulas and scoring rules |
| **2. Global Risk and Scenario Manual** | Full taxonomy, shock assumptions, transmission formulas, scenario selection and correlation treatment |
| **3. Data and Model Governance** | Sources, missing data, confidence, updates, version control, overrides and validation |
| **4. Worked Corporate Credit Assessment** | One complete issuer example with historical inputs, base score, stress scenarios, repayment analysis and audit trail |

All published scores should also distinguish the issuer's creditworthiness from the specific bond's seniority, guarantees, collateral, contractual maturity and recovery prospects.

For methodological reference, the following published frameworks are particularly relevant:

- Published corporate rating methodologies of the major credit rating agencies
- Their published management and governance criteria

These are methodological references, not endorsements of LR or evidence that the proposed LR weights have been validated.

# 10. Key design decision

I would not publish a final corporate LR formula yet. We now have a substantially more complete risk universe, but three technical questions must be resolved before freezing the methodology:

- Weights: Which risks belong in the 100-point base score, and which should affect only stressed outcomes?
- Quantification: How do we convert structural disruption, geopolitical exposure and key-person dependency into repeatable, evidence-based assessments?
- Aggregation: Should the final LR score reflect expected outcomes, downside resilience or a combination, and how do we avoid double-counting correlated risks?

My proposed solution is to preserve LR Base as a transparent 0–100 credit-strength score, publish LR Stress as a separate resilience profile, and develop a final composite only after historical calibration establishes defensible aggregation rules.

That separation would make the methodology more credible than attaching arbitrary numerical penalties to every conceivable event. It would also give 10bonds a useful investor-facing distinction: a company's current financial strength versus its ability to survive future disruptions.

The ambition should be an auditable, forward-looking framework whose added risk factors demonstrably improve credit assessment—not simply a longer checklist than the rating agencies use.